

Numerical

Given a data:

Predict - Relative Risk of Coronary Heart Disease (RR-CHD).

Patient	Systolic X_1	Diastolic X_2	Bmi X_3	Waist X_4	Actual Y
1	140	80	35	100	1.81
2	120	80	25	80	1.22
3	130	100	30	60	1.71

$$h_{\theta}(X) = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3 + \theta_4 X_4$$

The formula for Gradient Descent is:

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

For intercept, we are taking $x_0 = 1$

α for 5 parameters.

$$\theta = [\theta_0, \theta_1, \theta_2, \theta_3, \theta_4]$$

$$\alpha = 0.00001$$

Initially $\theta_0 = \theta_1 = \theta_2 = \theta_3 = \theta_4 = 0$.



$$h_{\theta}(x_i) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4 \dots$$

Patient 1)

$$h_1 = \theta_0 + 140\theta_1 + 80\theta_2 + 35\theta_3 + 100\theta_4$$

$$h_2 = \theta_0 + 120\theta_1 + 80\theta_2 + 25\theta_3 + 80\theta_4$$

$$h_3 = \theta_0 + 130\theta_1 + 100\theta_2 + 30\theta_3 + 60\theta_4$$

Iteration - I

$$\theta = \{0, 0, 0, 0\}$$

$$h_1 = 0, h_2 = 0, h_3 = 0$$

P	y	Predicted	Error P-y
1	1.81	0	-1.81
2	1.22	0	-1.22
3	1.71	0	-1.71

Calculate gradient θ_0 :

$$x_0 = 1.$$

$$1. \quad \frac{\partial J}{\partial \theta_0} = \frac{1}{3} (-1.81)(1) + (-1.22)(1) + (-1.71)(1)$$

$$\boxed{\frac{\partial J}{\partial \theta_0} = -1.58}$$

2. Calculate gradient for θ_1

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{3} (-1.81)(140) + (-1.22)(120) + (-1.71)(130)$$

$$\boxed{\frac{\partial J}{\partial \theta_1} = -207.36}$$

$$3. \quad \frac{\partial J}{\partial \theta_2} = \frac{1}{3} (-1.81)(80) + (-1.22)(80) + (-1.71)(100)$$

$$\boxed{\frac{\partial J}{\partial \theta_2} = -137.8}$$

Similarly calculate for

$$\theta_3 \text{ \& } \theta_4$$

Now update Gradient

Descent

$$\theta_j^{\text{new}} = \theta_j^{\text{old}} - \alpha (\text{gradient})$$

$$\alpha = 0.00001$$

Update

$$\theta_0 = 0 - (0.00001) (-1.58)$$

$$\boxed{\theta_0 = 0.0000158}$$

Similarly Update

$$\theta_1 = 0.0020$$

$$\theta_2 = 0.0013$$

$$\theta_3 = 0.00018$$

$$\theta_4 = 0.0012$$

Now Calculate Predictions.
after iteration - 1.

$$h_1 = 0.0000158 + (0.00207)(140) \\ + (0.001378)(80) + (0.00048)(35) \\ + (0.06127)(100)$$

$$h_1 = 0.54457$$

Similarly we get h_2, h_3, h_4 .
Put in in table

Patient	Actual	Prediction	Error
1	1.81	0.544	-1.26
2	1.22	0.472	-0.75
3	1.71	0.498	-1.21

Now go for Iteration 2

Now we will calculate
again the gradient with
New class

Repeat :-

1. Calculate gradient for

θ_0
 θ_1
 θ_2
 θ_3
 θ_4

2. Then update θ
with formula

$$\theta_j^{\text{new}} = \theta_j^{\text{old}} - \alpha (\text{gradient})$$

→ Once you get new θ ,
put in formula in $h_0(x)$.

get predicted value.

$$h_1 =$$

$$h_2 =$$

$$h_3 =$$

→ Once you get new predicted
value calculate error.

See if it converge or not.
otherwise repeat the iteration.
again till convergence.

Check this.

Linear Regression.

- Simple
- Multiple
- Iterative.